

Rohan Bali

CONTACT INFORMATION

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Links: Website · GitHub · Google Scholar · ORCID

RESEARCH INTERESTS

My research is on evaluation methodology and out-of-distribution generalization for spatiotemporal vision. I work on input-design covariates that confound architectural comparisons in earth-observation models (channel count, temporal horizon, geographic distribution), and on whether observed differences hold up under sealed holdouts and zero-shot geographic transfer. Broader interests: robust computer vision, distribution-shift evaluation, calibrated uncertainty in deep learning.

EDUCATION

University of Massachusetts Dartmouth, North Dartmouth, Massachusetts, USA

Department of Computer & Information Science

M.S., Data Science, September 2024 – May 2026

Advisor: Dr. Donghui Yan

Thesis: *Multi-Horizon Urban Growth Prediction: A Comparative Deep Learning Study*

Focus: Distribution shift, evaluation methodology, and uncertainty quantification in spatiotemporal vision

Manipal University Jaipur, Jaipur, India

School of Computing and Information Technology

B.Tech., Computer and Communication Engineering, August 2017 – May 2021

Advisor: Dr. Somya Goyal

Final-year project: *Road Safety Analysis using Machine Learning*

HONORS AND AWARDS

GeoAI 2026 Oral Presentation, June 2026

National Talent Search Examination (NTSE) Scholar, Government of India, 2016

PUBLICATIONS

Bali, R. 2026. The Channel-Count Confound: A Continental Audit of Multi-Horizon Urban Growth Prediction. GeoAI 2026, Ghent, Belgium. **(Oral)** Also at SpatialDI 2026, Springer LNCS proceedings, ACM SIGSPATIAL China Chapter, Changsha, China. Sole author. Distilled from the M.S. thesis.

Setup. Audits 5-year vs. 10-year urban-growth prediction across 5,698 CONUS tiles at 250 m under matched evaluation, with a sealed 2020 holdout and zero-shot transfer to 20 Lagos tiles (63,840 pixels).

Channel-count isolation. Moving from 5yr to 10yr removes one epoch, cutting the input stack from 24 to 21 channels. Prior work conflates this covariate with horizon length. 94% of CNN’s Figure-of-Merit decline from 5yr to 10yr attributes to the channel reduction (delta-method 95% CI [62%, 100%]; conservative $\geq 84\%$ floor; channel effect $8\times$ the two-seed $\sigma = 0.007$ spread).

Recency asymmetry. ConvLSTM shows $5.4\times$ larger response to end-removal than to interior-removal. The effect is positional, not volumetric.

Uncertainty and methodology. MC Dropout calibration over 8.69M pixels from 1,352 inference runs yields Pearson $r = 0.983$ between predicted $\hat{\sigma}$ and tile-level error (rank reliable). Absolute magnitude is underestimated by $1.4\text{--}1.9\times$. MSE inverts architectural rankings on land-cover tasks under specific Figure-of-Merit conditions.

GeoAI 2026 short paper (Zenodo). · Code and benchmark repository.

IN PREPARATION

Bali, R. Cross-Region Source-Invariance in Earth Observation. Sole author. Follow-on to the channel-count audit. Trains one model per region and evaluates every model on every

region across 20 world regions, 8 prediction tasks, and several input representations. The test region and input representation explain almost all cross-region score variance while the training region explains almost none, and a parameter-free recent-past baseline matches or beats trained models on temporal tasks. Code and data repository.

Bali, R. UrbanFinance: Forecasting the Latent Land-Demand Residual at 100 m across Divergent Indian Urban Typologies. Sole author. Predicts the signed gap between economically expressed demand and EO-realized supply across five divergent Indian metros (Noida, Mumbai, Chandigarh, Bengaluru, Hyderabad), fusing free GHSL, night-lights, terrain, and NHB RESIDEX layers. The DSRF model beats SimVP, TAU, and PredRNN.V2 on Figure of Merit at $p < 0.05$ over 10 seeds, gives calibrated 95% prediction intervals, and under leave-one-city-out transfers to an unseen city better than SOTA trained on all five.

TECHNICAL METHODS

Spatiotemporal architectures: ConvLSTM, CNN, U-Net, sequence models (LSTM, GRU, Transformer) for forecasting.

Evaluation under distribution shift: matched-tile audits, sealed temporal holdouts, zero-shot geographic transfer, Figure-of-Merit, MSE/FoM rank-inversion analysis, channel-volume controls.

Uncertainty quantification: MC Dropout calibration at pixel scale, reliability diagrams, predictive variance under horizon shift.

Earth-observation data: Landsat, Sentinel, NLCD; continental-scale tiling, harmonization, and export on Google Earth Engine.

Stack: Python, PyTorch, TensorFlow, xarray, Rasterio, GeoPandas, GDAL, scikit-learn, NumPy, Pandas.

SELECTED PROFESSIONAL EXPERIENCE

Capgemini, Mumbai, India

Software Engineer

December 2021 – August 2023

Supported production infrastructure and job-scheduling workflows; built Python, PowerShell, and Bash automation for data and systems operations across cloud environments.

Upcred.ai, Bangalore, India

Machine Learning Researcher

June 2021 – December 2021

Designed Python and Beautiful Soup web-scraping pipelines and MongoDB schemas supplying training data for the team's machine learning models.

SERVICE AND VOLUNTEERING

Founding Team Member, The Parivartan Foundation

2023 – Present

Registered non-profit based in Shimla, Himachal Pradesh, India. Source and analyze public government datasets to identify community-level priorities and inform targeted outreach campaigns for local development and civic engagement.